Transformers & NLP Notes

1. Transformer

- Advanced version of NLP.

- Neural network model used for sequential data.

2. Problems in RNN/LSTM

- Long dependency issues

- Slow processing

- Not scalable

3. Advantages of Transformers

- Parallel processing

- Highly scalable (BERT, GPT)

- Captures long-range dependencies

4. Architecture

Input → Encoder → Context → Decoder → Output

5. Core Components

- Self-Attention

- Positional Encoding

- Multi-Head Attention

- Encoder-Decoder

- Embeddings

- Tokenization

- Feed Forward Neural Network

6. Self-Attention

- Uses Query (Q), Key (K), Value (V)

- Formula: Attention(Q,K,V) = softmax(QK^T / √dk)V

7. Positional Encoding

- Adds position information to tokens

8. Multi-Head Attention

- Splits attention into multiple heads

- Captures different patterns

9. Encoder-Decoder

- Helps align input and output sequences

10. Embeddings

- Converts text into vectors

- Captures semantic relationships

- Reduces dimensionality

- Supports transfer learning

11. Tokenization

- Splits text into tokens

Example: "I love learning transformers"

→ ["I", "love", "learning", "transformers"]

12. Feed Forward Network

- Processes output from attention layers

- Applied after embeddings and encoding

13. Applications using Transformers

- Sentiment Analysis

- Text Classification

- Question Answering

- Text Generation

- Named Entity Recognition

- Summarization

- Translation